

SULONG ZHOU

Applied AI, LLMs, GeoAI, Large-Scale Spatial Data Systems

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EDUCATION

University of Wisconsin-Madison

Madison, WI

Ph.D.: Environment & Resources

Sep 2015 – May 2021

Dissertation: Applied Machine Learning for Monitoring the Environment | [Link](#)

M.S.: Computer Science

Sep 2019 – May 2021

WORK EXPERIENCE

Texas A&M University

College Station, TX

Postdoc Researcher - LLMs for Disaster Response

Oct 2024 – Present

- Built a multimodal topic modeling pipeline using LLM embeddings and BERTopic to analyze **150K+** Reddit posts, improving topic coherence scores by **38%** for spatiotemporal disaster response analysis.
- Managed and optimized PostgreSQL/PostGIS ETL pipelines processing **1.5B+** nationwide longitudinal property records to support large-scale data analytics across multiple research groups.
- Engineered features across **200+** variables to train a regression model (adjusted $R^2 = 0.87$) to quantify wildfire impacts on housing prices and identify lagged effects.
- Led development of an urban digital twin prototype by building an AR-assisted geospatial mobile application for real-time data collection and interactive urban design workflows.

ArtEast - An online fitness platform

Madison, WI

Data Scientist

Jan 2024 – Oct 2024

- Developed a conversational AI agent using OpenAI Realtime API to automate customer support service and reduce customer response time by **70%**.
- Analyzed customer behavior to guide new service strategy, driving a **12%** uplift in quarterly revenue.

DataChat - A conversational business intelligence platform

Madison, WI

Senior Software Engineer

May 2021 – Aug 2023

- Collaborated on GenAI integration and cross-team A/B testing to increase conversion rates by **20%**.
- Optimized backend message queue and Python APIs to reduce inference latency by **15%**.
- Refactored React components to ensure reusability and improve rendering performance by **30%**.

SELECTED PUBLICATIONS

- **S. Zhou**, et al. "Revealing time-sensitive latent public health and emotional distress topics in Reddit discourse during the 2025 Southern California wildfires". International Journal of Disaster Risk Reduction: 106167, 2026. | [Link](#)
- **S. Zhou**, et al. "Advancing smart disaster response by leveraging social sensing and mobile technology." Geo-spatial Information Science: 1-19, 2025 | [Link](#)

- **S. Zhou**, et al. "A guided latent Dirichlet allocation approach to investigate real-time latent topics of Twitter data during Hurricane Laura". Journal of Information Science, 49(2): 465-479, 2023. | [Link](#)
- **S. Zhou**, et al. "Application of image segmentation in surface water extraction of freshwater lakes using radar data." ISPRS international journal of Geo-information 9.7: 424, 2020 | [Link](#)

OPEN SOURCE PROJECTS

End-to-End Language Model Training Framework

2026

- Processed 2,500 raw Common Crawl WET files into a pretraining corpus through FastText quality filtering and Locality Sensitive Hashing (LSH) deduplication, removing **87%** of low-quality or duplicate documents.
- Built and trained a **45M-parameter** Transformer LM from scratch on **2.73B** training tokens with byte-level BPE tokenization, RoPE, and SwiGLU; stabilized training through LayerNorm placement and reduced validation loss by **18%** relative to baseline through hyperparameter tuning
- Profiled and benchmarked Transformer layers; optimized memory and computational efficiency using mixed precision, activation checkpointing, PyTorch compile, and Triton-based FlashAttention2, reducing training step time by up to **3.7×** and activation memory usage by **~50%**.
- Applied chain-of-thought prompting and **GRPO**-based reinforcement learning (RL) to improve OLMo-2-1B mathematical reasoning on GSM8K; compared **on- and off-policy** variants, with **GSPO** achieving the best validation correctness and format rewards.

Spring-based Recommendation Web Platform | [Git](#)

2019 – 2023

- Developed a client-facing ticket management web application using content-based filtering to improve recommendation relevance based on geographic and user behavioral data.
- Built RESTful Java servlets and optimized Apache Tomcat backend services to improve request processing and system responsiveness.
- Scaled infrastructure via Amazon EC2 to support a throughput of **160 QPS**, validated through performance benchmarking with Apache JMeter.

SKILLS

Programming & Tools: Python, Java, SQL, React, Git, PyTorch, Linux, AWS, Google Colab

LLMs: Transformers, Fine-tuning, RAG, Prompt Engineering, Multimodal Inference, Hugging Face

Machine Learning: Topic modeling (BERTopic, LDA), Regression, Gradient Boosting, A/B Testing

HONORS AND AWARDS

Workshop Grant (2025 Texas A&M Institute of Data Science)

Honorable Award (2025 AI and Cities: An International Forum for Innovation and Collaboration)

Selected Poster (2025 UT System AI Symposium in Health Care)

Student Scholarship (ESRI Geodesign Summit)

Selected Work for GIS Solutions Expo (ESRI Federal GIS Conference)